

Chow

Technical / water-system voice

How to use this: not a verbatim script — nobody is expected to read from it word-for-word, and realistically no one will stick to a script anyway. Study it well enough to explain each point in your own words, improvise if you're cut off, and answer a follow-up without freezing. If a question lands outside your section, give the short version and hand off by name — every presenter's script and the underlying `FIELD-NOTES.md` are in the same GitHub repo.

Field-visit recap (read this first if you missed a visit)

Everything in the deck traces back to these visits — this is the shared ground truth every presenter should know, whether or not you were there.

Tatay's cacao farm, Batbat (Tuesday, Aug 26). A 2.5-hectare farm run by Tatay alone — his children work in the cities — 100% organic, and he has health problems that limit how much heavy labor he can do himself. His farm used to produce about 500 kg of cacao a year (sold at ₱75–100/kg); this season it's close to **zero**, because of pests and disease. His own estimate of the farm's potential if pests were solved: **2,000 kg/year** — a 4× recovery, the biggest single lever we found anywhere. Cacao flowers year-round here, so the pest pressure never gets an off-season. On water, his framing was specific: the constraint is water *pressure*, not volume — he can't even estimate how many sprinklers his source could feed — and he's open to drip irrigation and rainwater collection. Important attribution point: Tatay described the problems and constraints; the pilot plan, the tools, and every proposed solution came from our own investigations, not from him.

What the farmers and Muravah told us directly. Pest control is their #1 problem — ahead of typhoon, which everyone called "unpredictable" and would rather we not chase. Post-harvest losses run 80–100% from pests/disease across the network, and 60% of pods die from careless handling alone (worse under Mayon ashfall). Farmer income is ₱2,000–10,000/month — below Bicol's ₱13,624/month poverty line for a family of five — the average farmer is about 50, and there's no succession pipeline; Muravah copes by recruiting BU Guinubatan students. Two hard rules for any solution: **organic only** (no synthetic fertilizers or pesticides, ever) and **no AI/robotics** ("not there yet," "AI is biased according to the natives") — low-tech and human-in-the-loop is a values position for them, not a budget issue. More labor doesn't raise profit, because pest pressure caps yield no matter how much work goes in.

Muravah Foundation / Mayon Gold. The cacao NGO anchoring all of this: cacao profits fund typhoon-proof homes and scholarships, heirloom native cacao interplanted with Pili and coconut, and a production-cycle data study already running across 500–1,000 farmers — which is where our pilot would live.

The processing facility. Their own #1 problem is power outages, but the line that matters for us: they're "sometimes out of supply for cacao." Farm-level pest losses upstream directly destabilize the factory

downstream — fixing the farms fixes two problems at once.

The other farms (Aug 27–28, now Extras slides). Tatay Aldavis's rice/dairy farm school (a different "Tatay" — it's a Bikol honorific for an elder, not a name) controls pests with herbal sprays and ducks that eat snails — proof organic pest control works in Bicol; plus a bamboo farm, stingless-bee honey producers, the Zambales agrivoltaics site (the ₱10M funding benchmark), and a youth agritourism program.

Your slides

You're the technical voice — the Batbat numbers, the field tools, and the pilot-study slide are all yours, so you carry the deck's hardest data slide and its most concrete engineering slide back to back with only Ash's callout and Nancy's farm photos between them. The rest of the team doesn't need to go deep on this material.

Slide 8 — The numbers from Batbat (3 charts).

1. *Loss by cause, crop by crop* — a grouped bar chart. Each cause has three bars: our cacao numbers next to two other Philippine cash crops. - **Handling:** cacao loses **60%** of pods to careless handling (field notes); Philippine rice loses **16.5%** total post-harvest [5]; dragon fruit loses **35%** of output post-harvest [17]. - **Pests & disease:** cacao **80–100%** (bar to 80, whisker to 100 — say the range out loud); rice **24–41%**, average ~37% of yield (IRRI, tropical Asia) [18]; dragon fruit stem canker **60–80%** in severe cases [19]. - **Typhoon (bad year):** cacao ~**100%**; rice **up to 95%** of standing crop in the worst-hit regions post-Haiyan [20]; dragon fruit **100%** — growers report total production loss when a typhoon topples the posts the vines grow on (storm Mario: 300 posts, ~4 t) [21]. If pressed on the dragon fruit figure, be honest that it's grower-reported rather than a published statistic. - The takeaway to land: even against crops with famously bad loss profiles, **cacao's pest/disease bar is the outlier** — worse than rice's worst case and above dragon fruit's most destructive disease.
2. *Tatay's cacao yield:* the chart shows all three bars — **500 kg/year** in previous years, near **0 kg** this season due to pest/disease — his estimated potential if pests were resolved is **2,000 kg/year**. Say this slowly: that's a **4× recovery**, and it's the single largest lever in the entire dataset.
3. *Farmer monthly income benchmarks:* a ₱2,000 low reference, **Blandino Papina** at ₱5,000 as the named example farmer, and ₱10,000 named as a "low" ceiling. The dashed line is Bicol's **poverty threshold for a family of five: ₱13,624/month** [4] — even the ₱10,000 "ceiling" sits ₱3,600 below it. Context for how thin margins are region-wide, not just for Tatay. The footnote adds broad context: PH fruits and vegetables lose 15–35% post-harvest [6], and cacao is far beyond that.

Slide 11 — Run the pilot through a live, calculated tool.

Explain what you built and why: rather than eyeballing a drip-irrigation pilot for Tatay's farm, you built a working set of field tools —

- a **drip-pressure planner** sized to his real 2.5-hectare / 120-meter layout,

- a **Raspberry Pi controller schematic** — the automation block that opens the zone valves (and the pump) on a schedule set by weather and season, from soil-moisture, rain, temperature and tank-level sensors, and
- an **interactive schematic editor**

(There's also a full **water-system schematic** — gravity vs. pump modes — reachable from the homepage and cross-linked from the controller page; it isn't a card on the slide, but mention it if someone asks how the pieces connect.)

How to read the water-system schematic, if someone opens it or asks. Water flows left to right: source → pump (only when the source sits below the field — Tatay's case; a gravity-fed source needs no pump, and that's the *only* component that changes) → check valve (one-way gate so water can't drain back toward the source) → filter (drip emitters clog without it — the #1 maintenance failure in real drip systems) → pressure regulator (steps pressure down to what thin drip line tolerates) → mainline splitting into **three zones, opened one valve at a time**, so a small low-pressure source can still water the whole farm. The teal dashed line is the **buffer tank** merging in as dry-season backup; the F and P circles are a flow meter and pressure gauge whose readings now also feed the Pi (pressure dropping = leak, flow dropping = clog — the Pi alerts the farmer by buzzer/SMS instead of anyone patrolling 2.5 ha of pipe, which answers the manpower and health constraint); the amber dashed lines are **control signals, not water** — the Raspberry Pi opening zone valves (and the pump, in pump mode) on a schedule it adapts to weather and season, with rain and soil-moisture sensors able to veto a run and the tank-level sensor keeping the pump from running dry. One-liner for the stage: *"source → lift if we need it → clean it → calm it down → one zone at a time out to the trees, with a rain tank as backup and a Pi deciding when."*

How the pressure planner calculates, if someone asks for the math. Five chained steps:

1. **Trees** = farm area ÷ spacing² — at the DA guide's 3×3 m spacing that's ~1,111 trees/ha, so ~2,778 on Tatay's 2.5 ha [3].
2. **Flow demand** = trees × emitters per tree × emitter flow rate — with 2 emitters at 4 L/hr each, Tatay's whole farm at once would demand ~22,000 L/hr (~370 L/min). That one number is the problem: almost no small source supplies it, which is why zoning exists.
3. **Required pressure** = emitter working pressure (~1.0 bar for standard drip) + friction loss (pressure dies with pipe distance, worse in narrow pipe) – elevation assist (**0.0981 bar per metre of height** — source above the field helps for free, below the field adds a pump). If the source falls short, the shortfall ÷ 0.0981 converts directly into the booster-pump head in metres — the exact spec you'd hand a pump seller.
4. **Zones** = flow demand ÷ source flow, rounded up — water one section at a time in sequence, the same one-valve-at-a-time logic on the schematic, automated by the Pi.

5. **Buffer tank** = daily watering volume × dry days to bridge, ÷1,000 = how many standard 1,000 L tanks — the two tanks in the ₱95K ledger come from here. One-liner for the stage: *"spacing tells you trees, trees tell you flow, distance and height tell you pressure; too little flow means zones, too little pressure means a pump, and the dry days you want to survive size the tank."* Water-needs grounding: the cocoa water-relations literature and FAO-56 [7][8].

— so any water-resilience pilot for Tatay is calculated from his actual numbers, not guessed. The water math itself is cited on the slide ([3]–[8]: the DA cacao production guide, the cocoa water-relations literature, FAO-56). Point the room at the GitHub repo (github.com/Abdullah-Nasir-Chowdhury/cacao-bicol-field-tools), which also holds the full field-notes reference and the interactive chart dashboard this deck draws its numbers from. Keep the framing tight: this is the concrete, checkable, technical contribution — invite people to open it later rather than trying to demo it live if time is short.

Slide 12 — The pilot study: pest and disease control.

You keep the floor straight after the tools slide — this is the substance of the pilot your tools make calculable: two organic lines of defense, both with published research behind them (the citations are clickable on the slide):

1. **Pest control — deploy insects that eat the pests.** We release beneficial predators against the snails and other pests: carnivorous snails that hunt the pest snails, plus red weaver ants and lynx spiders. This isn't speculative: DOST-PCAARRD is **already mass-rearing weaver ants and lynx spiders** as biological control agents for cacao insect pests in the Philippines, and field research shows weaver ants cut pest damage in cacao [15]. If someone asks "does this fit organic?" — yes, biological control is the textbook organic answer, and it echoes what the rice/dairy farm already does with ducks eating snails.
2. **Disease control — keep the trees watered and healthy.** Water stress measurably raises a plant's susceptibility to disease, and the classic plant-pathology finding (Schoeneweiss 1975) is that the effect **reverses once regular watering resumes** [16]. That's the scientific link between the water system and the disease problem: the drip system plus the Pi controller keeps trees out of the stress zone year-round, which is disease prevention, not just irrigation. Land the closing line: both measures satisfy Muravah's hard rule — no synthetic pesticides.